Hilbert-Filled Inference Spaces for Automated Theorem Proving

GBG+ with Player-0 = $B^\star$: Operational Search, Empowered Theorems, and Research Frontiers

Integrated Research Edition v2.0 — ATLAS Mathematical Repair

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See the AI disclaimer that follows this essay.

Abstract

This research edition retains the original program of Hilbert-directed theorem search, its exact-rank and rational-bucket geometries, an underlying generalized board game (GBG), and the passive and actively controlled GBG+ extension with Player-0 $B^\star$. It repairs several logical boundaries. The finite Hilbert covering theorem concerns a frozen finite candidate set; target-search breadth layers preserve *relative rule depth*, not an unspecified total proof length. Equivalence between theoremhood and winning-state reachability requires fair admission of every proof action, and cannot be concluded merely from a finite epoch hypothesis that excludes infinite first-order axiom schemas. Passive board-player cancellation does not authorize active mutation of logical resources. A reported local first-order set-theory prototype is distinguished from presently reproduced evidence: its original companion code and proof receipts were not supplied in this revision. Our proofs are ordinary mathematical arguments, and neither ZFC completeness nor an unconditional Hilbert search advantage is claimed.

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1 The executable claim and its limitations

The central proposal is operational rather than metaphorical: finite d-dimensional Hilbert traversals can schedule candidate applications of d-premise inference rules. Exact WFF identity and verifier acceptance remain separate from the navigation coordinates. The intended pipeline is

$$\text{syntax} \longrightarrow \text{premise tuples} \longrightarrow \text{Hilbert cells} \longrightarrow \text{rule attempts} \longrightarrow \text{checked deposits.}$$

Traversal is a permutation of a finite action list; it does not create inference legality or change the final set of conclusions of a fully completed frozen stage. The open scientific question is whether a useful geometric address map reduces *charged, finite-budget* discovery cost against strong baselines.

Throughout, a *verifier-accepted* formula means that its given finite rule-and-premise record passes a declared checker. This does not, by itself, assert that the checker implementation has been independently verified. A proof may be valid relative to a declared theory without constituting a new theorem of the bare logical calculus.

2 The syntactic backbone

Let $F = (x, y, z)$ be a formal system: x is an alphabet, y its set of WFFs, and z a rule stock with concrete finitary instance descriptions and side conditions. We distinguish object-language formulas from machine states, candidate rule attempts, and proof certificates.

Proposition 2.1 (Effective WFF indexing)**.** *Assume that y is infinite, WFF membership among finite strings is decidable, and the string language has an effective injective coding by natural numbers. There is an effective bijective enumeration $n \mapsto \varphi_n$ of y.*

Proof. Scan natural codes in order. For each code, decode when possible and retain exactly valid WFFs. Injectivity prevents repeats. Since y is infinite and membership decidable, the nth retained WFF is located after finitely many scans, and every WFF eventually appears. □

This index is an exact syntactic identifier, not a Hilbert coordinate. A separate mapping $\lambda : y \to \mathbb{Q} \cap [0, 1)$ may be used as navigation metadata. For each frozen finite $C \subseteq y$ an injective rational address map exists trivially; proposed semantic locality is an additional hypothesis for experimental study.

3 Rule arity determines candidate dimension

For a frozen finite trusted set C and a concrete rule tag r_i of arity $d_i \geq 0$, its explicit-premise candidate family is $\{r_i\} \times C^{d_i}$. For a finite tagged rule library define

$$\mathcal{A}_F(C) = \coprod_{i=1}^{m} \{r_i\} \times C^{d_i}, \qquad C^0 = \{()\}. \tag{1}$$

Tags prevent accidental identification of different rules. When a rule has additional substitutions or schematic parameters, those parameters are also part of an *action description*; Equation (1) is the whole action space only when those extra data are fixed or encoded in the tag. The arity of an inference belongs to its rule application, not to an individual WFF. Zero-ary rule *families* with infinitely many axiom instances cannot be silently treated as a finite candidate set.

3.1 Exact-rank cells

Write a frozen nonempty list $C = (\theta_0, \ldots, \theta_{M-1})$, and define

$$p = \lceil \log_2 M \rceil, \quad N = 2^p \quad (M = 1 \implies N = 1).$$

Let $H_{d,p} : \{0, \ldots, N^d - 1\} \to \{0, \ldots, N - 1\}^d$ be a chosen bijective discrete Hilbert-style traversal. The exact-rank cell of a tuple $(\theta_{j_1}, \ldots, \theta_{j_d})$ is $(j_1, \ldots, j_d)$. We distinguish the existence of a bijection from additional geometric conventions such as face adjacency.

Theorem 3.1 (Exact finite Hilbert coverage)**.** *For $M \geq 1$ and $d \geq 1$, a complete pass through $H_{d,p}$ encounters every ordered tuple in C^d exactly once among its valid (nonpadding) cells.*

Proof. Each valid tuple gives the unique index vector $(j_1, \ldots, j_d) \in \{0, \ldots, M-1\}^d$. The traversal is bijective on the larger padded grid, so this vector occurs exactly once. Grid cells with an index at least M are padding and are ignored. □

If $C = \varnothing$ and $d \geq 1$, the candidate family is empty and the traversal is skipped. For $d = 0$ there is a singleton empty premise tuple, to be handled separately from geometric coverage. Exact coverage counts M^d represented candidates, not their unknown cost, usefulness, or legality.

3.2 Rational addresses and finite buckets

Fix an injective rational code $\lambda : C \hookrightarrow \mathbb{Q} \cap [0, 1)$. For a resolution $q \geq 0$, put $L = 2^q$ and

$$J_q(u) = \lfloor Lu \rfloor, \qquad B_j = \{\varphi \in C : J_q(\lambda(\varphi)) = j\}.$$

Although λ is injective, $J_q \circ \lambda$ need not be. A cell $(j_1, \ldots, j_d)$ thus stores a finite Cartesian bucket product $B_{j_1} \times \cdots \times B_{j_d}$.

Theorem 3.2 (Bucketed finite Hilbert coverage)**.** *If $H_{d,q}$ bijectively traverses the L^d hypercells and each bucket product is enumerated without duplicates, a completed pass enumerates precisely C^d, once per ordered tuple.*

Proof. Each $\varphi \in C$ has exactly one quantized index $J_q(\lambda(\varphi))$. Thus each tuple lies in exactly one bucket product. The traversal visits that product once, and its stipulated local enumeration visits each member once. □

The rational realization is intended to support a locality hypothesis. Neither injectivity before quantization nor Hilbert continuity alone proves a performance improvement.

4 Hilbert breadth-first target search

4.1 Finite relative search universe and proof-depth metric

Let $Y \subseteq y$ be a *finite* represented WFF universe, $\Gamma \subseteq Y$, and $T \in Y$. Define the direct-consequence operator of represented rules restricted to Y by

$$D_{F,Y}(C) = C \cup \{\psi \in Y : \psi \text{ is the checked conclusion of a represented legal rule instance over } C\}.$$

All zero-premise instances counted at a stage must be included in the declared finite candidate list, or introduced by a specified fair stagewise enumeration. Fix completed breadth sets

$$C_0 = \Gamma, \qquad C_{m+1} = D_{F,Y}(C_m). \tag{2}$$

Define a derivation's *rule depth* as 0 for initial hypotheses, and one plus the maximum premise depth for a noninitial inference. This is not necessarily the total number of rule occurrences or the size of a serialized proof DAG. Let $h_T^Y(\Gamma)$ be the minimum rule depth of a proof whose intermediate formulas all lie in Y using the represented rules, with value ∞ if none exists.

The restricted horizon h_T^Y equals the unrestricted depth h_T only if Y contains a minimum-depth proof of T or a separate adequacy theorem is established. Finite Y alone does *not* establish that $\Gamma \vdash_F T$ implies discovery.

4.2 Frozen-layer Hilbert scheduling

At stage m freeze the completed C_m and choose a schedule-independent canonical ordering of its WFFs. For each rule tag of arity $d_i \geq 1$, assign the Hilbert key $\eta_{i,m}(j_1, \ldots, j_{d_i}) = H^{-1}_{d_i,p_m}(j_1, \ldots, j_{d_i})$, $p_m = \lceil \log_2 |C_m| \rceil$, with the nonempty-set conventions above. Zero-ary tagged actions use a fixed independent order.

Definition 4.1 (Hilbert breadth-first policy (HBF))**.** HBF freezes C_m, enumerates *all* represented candidate actions required for $D_{F,Y}(C_m)$, orders premise tuples by Hilbert key within a fixed rule-tag order, tests actions against the declared verifier, commits all verified new conclusions simultaneously, and freezes after the first *completed* layer containing T. Memoized results may be reused only when the frozen inputs and exact action semantics are unchanged.

Theorem 4.2 (Completed-layer invariance and relative horizon)**.** *Assume each frozen layer has a finite completely enumerable candidate list, each candidate's outcome is determined only by frozen inputs and its action description, and the verifier is exact for the represented rule instances. HBF and a lexicographic complete breadth scheduler have the same C_m at every completed stage. The first target-containing stage, if any, is exactly $h^Y_T(\Gamma)$.*

Proof. At stage zero both have $C_0 = \Gamma$. If their current sets agree, the two schedulers operate on the same finite multiset of candidate applications; order is the only difference. Identical verification and atomic commitment produce identical C_{m+1}. Induction proves completed-layer equality. Another induction on the definition of rule depth proves that C_m consists of exactly the formulas with represented proof depth at most m. This yields the horizon equality. □

Corollary 4.3 (Early exit has minimum *relative depth*)**.** *If T is obtained at a microstep during construction of the first target-containing layer, and every preceding layer was completed, immediate exit provides a proof of minimum depth $h^Y_T(\Gamma)$. Its partial final ledger is* not *the completed breadth-layer state. The statement does not claim minimum number of steps or minimum cost.*

Proof. The successful candidate has premises in C_{m-1}, hence produces a depth-m proof. Had a smaller-depth proof existed in the represented universe, T would have belonged to an earlier completed layer, contrary to hypothesis. □

4.3 Within-layer discovery and the controlled experiment

For scheduler σ, $\kappa_\sigma(\Gamma, T)$ is the index of the first candidate attempt producing T in its first target layer, and E_σ counts all candidate attempts through that event, including preceding layers. They may differ across schedules despite the invariance of completed layers and target depth. Record completed candidate counts, attempted and verified actions, proof certificates, time, peak memory, receipt storage, and communication cost. Benchmarks should compare **BF-Lex**, seeded **BF-Rand**, and **HBF**, keeping theory, Y, Γ, T, rule-tag order, parser, checker, hardware, and resource budgets fixed.

The supplied edition 1.0 reported the following historical toy comparison. The underlying executable files and raw receipts were *not attached* to the present audit, so these figures are source claims and have *not* been independently rerun here.

Target	Policy	Depth	Reported microstep	Completed attempts
7	Lex	3	5	30
7	Hilbert	3	8	30
7	Random	3	7	30
13	Lex	3	24	30
13	Hilbert	3	20	30
13	Random	3	17	30
31	Lex	4	54	210
31	Hilbert	4	76	210
31	Random	4	52	210

As reported, the example illustrates schedule-dependent discovery position with identical completed-layer counts; in the sample Hilbert can be better or worse. It is not a measured overall speedup.

5 Search-state lifting to an underlying GBG

Definition 5.1 (Complete ATP search state)**.** An encoded state S stores all information that may influence future legality or verification: trusted formulas C, proof records P, candidate queues and cursors, epoch and Hilbert tick, declared policy, remaining budgets, import receipts, scope, and authorization metadata. Write $C(S)$ for the trusted component.

Theorem 5.2 (Search-state lifting)**.** *If legal complete states have finite effective encodings and each executable elementary action has a finite label containing all nondeterministic choices needed to make its successor unique, then the ATP transition graph has a generalized board-game representation with valid state encodings as positions and partial unary action functions $R_\mu : S \rightharpoonup S$ as legal moves.*

Proof. Encode each complete state as a finite string and each choice-resolved action by a finite label μ. Let R_μ be defined exactly on states where the action is legal, with the encoded successor as its value. An actual finite execution is now exactly a finite path in the directed labeled transition graph, and conversely. An active board-player extension requires further declared, authorized moves; none is inferred from this encoding alone. □

This separates (a) the underlying GBG geometry of *complete operational states* from (b) the Hilbert geometry of *candidate rule applications*. A Hilbert tick may guide the next GBG move, but it is not itself an additional inference rule.

6 Target winning region and soundness

For target T define the target region $W_T = \{S : T \in C(S)\}$. Write $\mathrm{Reach}(S_0)$ for states reachable by finitely many legal actions from initial state S_0.

Theorem 6.1 (Target reachability as an operational identity)**.** *There exists a finite legal run successfully depositing T if and only if $W_T \cap \mathrm{Reach}(S_0) \neq \varnothing$.*

Proof. A successful run is a finite path ending in a state containing T, hence supplies a member of the intersection. A member of the intersection is by definition a legally reachable state containing T. □

Corollary 6.2 (Certificate-gated soundness)**.** *If every trusted insertion is supported by a finite verifier-approved proof step in F whose premises are already trusted, and initial trusted data have authorized support Γ, then $W_T \cap \mathrm{Reach}(S_0) \neq \varnothing$ implies $T \in \mathrm{Con}_F(\Gamma)$.*

Proof. Trace the finite provenance of the deposited target backwards. Every derived leaf eventually reaches a declared axiom or authorized hypothesis, yielding a finite F-derivation of T. □

Existential reachability must not be confused with what a particular policy will actually reach. A controller can legally choose a dead end even when a winning path exists.

7 Provability and fair eventual discovery

7.1 The finite-complete-epoch theorem, properly restricted

Theorem 7.1 (Finite complete epochs)**.** *Assume $C_0 = \Gamma$ is finite; the represented rule library has a finite complete list of actions over each finite C_e (including all relevant axiom actions); every accepted action is sound; and every completed epoch computes precisely the full direct-consequence set $C_{e+1} = D_F(C_e)$. If the system continues until success or a genuine fixed point, then*

$$T \in \mathrm{Con}_F(\Gamma) \iff \textit{some finite complete epoch deposits } T,$$

for targets whose proofs use the represented rule instances and whose direct-consequence operator is exactly that of F.

Proof. A finite proof of T has finite maximum rule depth r. Induction over its proof lines and the complete direct-consequence property places each line of depth j into C_j, so $T \in C_r$. Each of the first r epochs finishes in finite time by the action-finiteness hypothesis. The converse follows from certificate-gated soundness. □

Important limitation. Many first-order theories have infinitely many logical axiom instances or substitutions even when C_e is finite. They therefore do *not* satisfy the literal "all applicable actions form a finite list" condition without an additional finite-stage enumeration device. The existence of a proof cannot justify this hidden finiteness assumption.

7.2 Fair dovetailing without finite global epoch closure

Theorem 7.2 (Finite discovery under fair admissible-action enumeration)**.** *Fix a sound, effectively enumerable finitary proof system and a finite authorized hypothesis bank Γ whose members are all initially admitted (or, more generally, are each fairly admitted at some finite time). Suppose the search produces a monotone verified ledger and obeys the following fairness condition: whenever an instance of a legal finitary rule has all its premises permanently in the ledger, that particular instance is attempted and, if valid, accepted at some finite later time. Suppose the machine takes infinitely many finite steps unless it has deposited T, with no permanent cutoff or starvation. Then*

$$T \in \mathrm{Con}_F(\Gamma) \Longrightarrow T \textit{ is accepted at a finite time.}$$

If accepted deposits are sound, the converse holds as well.

Proof. Choose one finite proof of T. Hypothesis and fixed axiom leaves enter in finite time by the assumptions (axiom enumeration is covered by zero-premise fairness). Induct on proof depth. Once a line's finitely many premises have entered, fairness attempts the particular rule instance deriving that line at some finite later time; acceptance adds it. Since the proof has finitely many lines, their last admission time is finite, and the final line T is eventually admitted. Soundness supplies the reverse implication. □

The theorem says nothing about a computable uniform *time bound* and does not decide non-theoremhood. It is a semi-decision result under a strongly fair ongoing controller, not a claim that a finite practical budget can settle every conjecture.

8 Expanding premise-action boards

For a finite library of explicitly parameterized rules, let

$$\mathcal{B}_e = \coprod_i \{r_i\} \times C_e^{d_i}, \qquad C_e \subseteq C_{e+1}.$$

Theorem 8.1 (Expanding board chain)**.** *If the rule tags and action semantics are fixed, each C_e is finite, and every r_i has finite arity, then every $\mathcal{B}_e$ is finite, the maps $\mathcal{B}_e \hookrightarrow \mathcal{B}_{e+1}$ are canonical, and*

$$\bigcup_{e<\omega} \mathcal{B}_e \cong \coprod_i \{r_i\} \times C_\infty^{d_i}, \quad C_\infty = \bigcup_{e<\omega} C_e.$$

Proof. Finite products and finite coproducts of finite sets are finite. Monotonicity gives inclusions. A tuple in the right side has finitely many entries, each appearing by a finite stage; choose the maximum of those stages. The tuple is then already present in the corresponding $\mathcal{B}_e$. The reverse inclusion is immediate. □

If a rule has a separately unbounded substitution parameter, it must be included explicitly in an expanded enumeration and cannot be erased from the finiteness hypothesis. A legal expanding board also does not authorize a Player-0 action that changes the logic itself.

9 Synchronized two-dimensional Hilbert atlas

Choose bijective finite Hilbert paths

$$H_{d,q} : \{0, \dots, 2^{qd} - 1\} \to \{0, \dots, 2^q - 1\}^d,$$

and suppose qd is even. Write $s = qd/2$ and choose the two-dimensional path $H_{2,s}$ on a square grid with the same number of cells. Define

$$\mathcal{A}_{d,q} := H_{d,q} \circ H_{2,s}^{-1}.$$

Theorem 9.1 (Path-order synchronization)**.** *$\mathcal{A}_{d,q}$ is a bijection of the two grid-cell sets, and for every shared tick t,*

$$\mathcal{A}_{d,q}(H_{2,s}(t)) = H_{d,q}(t).$$

If both chosen paths have face-adjacent consecutive cells, then consecutive path ticks *are face-adjacent in each respective grid.*

Proof. Both Hilbert paths are bijections from index sets of size 2^{qd}; their composition is a bijection. Substituting $H_{2,s}(t)$ gives the identity. The adjacency statement is exactly the independently assumed face adjacency of each path, not a metric statement about arbitrary pairs of grid points. □

When qd is odd, an exact square two-dimensional grid of the specified form does not have matching cardinality; use padding, a rectangular raster, or changed resolution, and state any loss of exact square-path equivalence. This atlas is an order synchronization, not a theorem that two-dimensional Euclidean neighborhoods represent proof-semantic neighborhoods.

10 Scheduler invariance and the finite-budget question

Proposition 10.1 (Invariance after completed frozen actions)**.** *Two schedulers visiting exactly the same finite frozen candidate multiset with identical verifier semantics and no within-stage feedback produce the same accepted conclusions after completion.*

Proof. The verified outcome of each action is independent of its position in the permutation. The final set is the union of the same accepted outcomes. □

For an interruption budget B, the prefixes $P_B(\sigma) = \{\sigma(1), \dots, \sigma(B)\}$ can differ, and one may encounter the target sooner. This is the only intended source of a geometric scheduling advantage in the frozen-layer comparison. Proving a useful advantage requires a distribution of targets or an informed policy class, cost accounting, and a fair comparator.

Part I
GBG+ Player-0 research extension

11 Player-0, board data, and authorizations

The original source adopts the notation $P_0 = B^\star$ for the added board-player participant in a GBG+ structure $G^+ = (G, H, \mathrm{Auth}^+, B^\star, \mathsf{F})$, with the label map $\mathsf{F}(r) = \max(0, r-1)$ in the cited N19 specialization. The *board* B or candidate action board $\mathcal{B}_e$ is *data*; $B^\star$ is a formal player label. The map F acts on player labels and by itself is not an inference rule or a scheduling policy. The present revision does not assume more about the unavailable N19 implementation than the explicit passive construction below.

A complete active-control state may include

$$S^+ = (C, P, e, h, Q, \beta, \sigma_0, \alpha, V, K_0, L, \Pi),$$

with verified local formulas C, provenance P, epoch e, tick h, queues Q, budget β, policy σ_0, authorization α, checker V, frozen input BANK K_0, provisional accepted ledger L, and receipt custody Π. Every operational coordinate that can change future transitions is part of the state. This structure *proposes* legal control packages; the board-player does not acquire permission to alter axiom stocks, hypothesis scope, or verifier standards implicitly.

12 The passive extension: cancellation, reachability, and history counts

12.1 An explicit passive construction

To avoid relying on an unstated external convention, define $E(G)$ here. Each underlying state s has tagged copies $\langle s, 0\rangle$ and $\langle s, 1\rangle$. Every underlying authorized move $\mu : s \to s'$ lifts to

$$\langle s, b\rangle \xrightarrow{\mu} \langle s', 0\rangle \quad (b = 0, 1).$$

An additional *passive* administrative action ε sends $\langle s, 0\rangle \to \langle s, 1\rangle$ only when at least one underlying ordinary move is currently legal. It changes no underlying state component, receipt, or theorem. The initial state is $\langle s_0, 0\rangle$. Let U erase the bit and administrative steps, retaining the ordinary moves and their successors.

Theorem 12.1 (Passive cancellation)**.** *For this construction, $U(E(G)) = G$ as the designated underlying labeled game, and for the canonical ordinary-history lift E_H, $U(E_H(\pi)) = \pi$ for each finite ordinary history π. These are identities of the* specified passive *construction, not of arbitrary active GBG+ policies.*

Proof. Each ordinary move has exactly the underlying source and successor it had before extension. Deleting the tag and ε returns the original states and ordinary move relation. Lifting a history using only bit-zero tagged states and ordinary moves then erasing the bit returns the original history verbatim. □

Corollary 12.2 (Passive target projection)**.** *Set $\widehat{W}_T = \{\langle s, b\rangle : s \in W_T, b \in \{0, 1\}\}$. Then*

$$\mathrm{Reach}_G(s_0) \cap W_T \neq \varnothing \iff \mathrm{Reach}_{E(G)}(\langle s_0, 0\rangle) \cap \widehat{W}_T \neq \varnothing.$$

Proof. Lift an ordinary winning history without passive steps. Conversely erase ε steps and tags from a winning enlarged history; ordinary moves remain legal and terminal membership is unchanged by the bit. □

Proposition 12.3 (Passive history multiplicity with fixed endpoint)**.** *Fix an ordinary legal history containing exactly n ordinary moves. Among enlarged histories that project to it and* end immediately after its nth ordinary move*, there are exactly 2^n distinct passive lifts, including the empty history when $n = 0$.*

Proof. Before each of the specified ordinary moves the bit is zero. There is exactly one optional ε insertion, which is legal because the specified ordinary move is enabled. The ordinary move resets the bit to zero. Independent binary choices at the n positions yield 2^n lifts. The endpoint convention excludes trailing administrative steps. □

13 Empowered theorems: an explicit preservation map

Here "empowered" means an original HFIS statement is restated with specified passive projection or legal Player-0 authorization; it does *not* necessarily mean a novel independent mathematical theorem.

Original claim	GBG+ interpretation	Essential hypothesis

WFF indexing	Preserve syntactic WFF identity	Frozen parser/coding
Exact/bucket coverage	Player-0 chooses a complete cell traversal	All candidate buckets enumerated
Breadth-layer invariance	Authorized within-layer permutation	Same finite frozen action multiset
Search-state lifting	Player-indexed transitions on full state	Legal action labels and custody
Target reachability	Existential winning region	Verified deposits
Eventual discovery	Policy-qualified fair search	No starvation or cutoff
Expanding boards	Monotone legal finite action boards	Frozen rule identity and scopes
2D atlas	Shared Hilbert tick across views	Equal cardinality and path convention
Scheduler invariance	No logical gain from permutation alone	Completed common candidate set

14 Policy-qualified discovery and counterexample

Theorem 14.1 (Actual controller success under fairness)**.** *Let π_0 be an authorized Player-0 search policy whose run from S_0 satisfies the fairness and sound deposit hypotheses of Theorem 7.2, with no permanent budget exhaustion or indefinite suspension. Then*

$$T \in \mathrm{Con}_F(\Gamma) \Longrightarrow \exists n < \omega : \mathrm{Run}_{\pi_0}(n) \in W_T.$$

A certified winning deposit conversely entails $T \in \mathrm{Con}_F(\Gamma)$.

Proof. Apply Theorem 7.2 to the actual fair run; use certificate-gated soundness for the converse. □

Proposition 14.2 (Existential reachability is weaker than policy success)**.** *A finite legal winning path can exist while a legal deterministic controller does not reach any winning state.*

Proof. From an initial state give the controller a choice between move a to a certified target state and move b to a non-target terminal dead end. The policy choosing b fails despite existence of winning path a. □

15 No universally best fixed order and the budget boundary

Theorem 15.1 (Target-agnostic non-dominance)**.** *For two distinct permutations $\sigma \neq \tau$ of the same finite candidate set, if any single action may be the unique target-producing action, neither permutation has a weakly smaller discovery index for every possible unique target action.*

Proof. Write $r_\sigma(a)$ and $r_\tau(a)$ for each candidate's absolute rank. As the permutations contain exactly the same n items, $\sum_a r_\sigma(a) = \sum_a r_\tau(a) = n(n+1)/2$. Because the permutations differ, their rank functions are not identical. Consequently at least one a has $r_\sigma(a) < r_\tau(a)$, and at least one b has $r_\sigma(b) > r_\tau(b)$; otherwise equality of the sums would force identical ranks. Make a, respectively b, the unique successful action. Each schedule is strictly earlier in one admissible assignment. □

The repaired argument uses absolute ranks, not merely inversion of the *relative* order of two items. The theorem is adversarial: it does not rule out a Bayesian or learned policy with better expected cost on a restricted target distribution.

16 Future-complete state quotients and verifier custody

Theorem 16.1 (Future-complete endpoint equivalence)**.** *Suppose every legal action, verifier response, and successor state is determined by the current* complete *state and its declared action label. If two histories end in exactly the same complete state, they have the same sets of finite legal labeled continuations and possible future certified deposits.*

Proof. Legal next actions and successors coincide by the full-state transition contract. Induct on continuation length. Deposits are transition outputs and therefore coincide for the same labeled continuation. □

Proposition 16.2 (Theorem-store equality is inadequate)**.** *Two states can have the same trusted theorem set and distinct legal futures.*

Proof. Take identical trusted formulas but candidate-attempt budgets $\beta = 1$ and $\beta = 0$. A new proof attempt requiring one budget unit is enabled only in the former state. □

Proposition 16.3 (Conditional local kernel soundness)**.** *Suppose every starting axiom is logically valid or a consequence of the declared theory, every assumption is retained in the official support, and every accepted inference is a correct capture-avoiding instance of universal elimination, MP, conjunction elimination, equality reflexivity, or equality symmetry. Then every accepted proof-record formula follows from its declared hypotheses and axioms.*

Proof. Induct over the finite proof record. The starting cases are authorized. Each permitted rule is sound relative to the same support, so every later accepted formula remains a consequence. □

This conditional proof of logical soundness is not a formal verification of any Python implementation. In particular a typed but incorrect parser, substitution routine, or receipt verifier could fail to meet the hypothesis.

17 Certified delegation, BANK conservativity, and full costs

Proposition 17.1 (Conservative checked imports)**.** *Fix an unchanged calculus F and an authorized bank Γ. For any $L \subseteq \mathrm{Con}_F(\Gamma)$ of correctly typed, independently proved lemmas,*

$$\mathrm{Con}_F(\Gamma \cup L) = \mathrm{Con}_F(\Gamma).$$

Proof. Monotonicity gives the forward inclusion. A derivation from $\Gamma \cup L$ uses finitely many imported lemmas, each individually derivable from Γ. Substitute their proofs into the derivation to obtain a Γ-derivation. Thus the reverse inclusion holds. □

This logical conservativity result does not install any AMLD+0 federation or certify distributed proof receipts. The input BANK K_0 must remain frozen; provisional ledger L may be grown under an explicit acceptance policy without silent writeback or hypothesis widening.

For a controller run define a resource vector

$$\mathbf{c} = (N_{\text{candidate}}, N_{\text{check}}, N_{\text{board}}, N_{\text{index}}, N_{\text{comm}}, M_{\text{peak}}, t_{\text{wall}}),$$

with nonnegative declared weights ω and $\text{Cost}_\omega = \omega \cdot \mathbf{c}$. Passive administrative transitions may have ordinary inference-ply cost zero while still consuming actual computational resources. No complete resource-cost comparison is available from the supplied edition's toy attempt counts.

18 Expanding boards under an authorized controller

Proposition 18.1 (Exhaustive finite-board visit)**.** *Let $\mathcal{A}_e$ be an increasing sequence of finite sets of legal candidate actions. If every action appears in some finite $\mathcal{A}_e$ and the declared controller completes each $\mathcal{A}_e$ in finite time without starvation, then every action is visited at finite time.*

Proof. Fix an action a and choose an e with $a \in \mathcal{A}_e$. By hypothesis the policy completes this board in finite time, and so must visit a. □

This proposition is about fair enumeration, not successful proof discovery unless the action list includes all steps of a needed proof with trusted premises. Atlas synchronization remains a display/order identity from Theorem 9.1, subject to declared board-player permissions.

19 First-order set-theory prototype: source evidence and boundaries

The source edition reports a limited deterministic implementation over the primitive first-order language $\{\in, =\}$ with logical connectives and quantifiers. It lists exact extensionality and empty-set sentences as a small trusted axiom stock, and reports local verification of MP, conjunction elimination, equality reflexivity/symmetry, and capture-checked universal elimination. These rules are sound *when correctly implemented*; they are not a complete prover for ZFC.

The reported local task was the *conditional* extensionality consequence

$$\forall z\big[(z \in x \to z \in y) \wedge (z \in y \to z \in x)\big] \;\vdash_{\text{Ext}}\; x = y.$$

The formula at left is an explicit hypothesis, not an unconditional set-theory theorem. Under ordinary extensionality it indeed implies $x = y$. The original text reports the following small experiment, but the claimed original scripts `src/`, `hilbert_bfs.py`, `check_receipt.py`, and receipts were *not among the three supplied PDFs*. Therefore no claim of reproduction, running ten tests, or inspecting a Python proof record is made in this revision.

Reported policy	Source outcome	Completed epochs	Attempts
Hilbert, exact rank	PROVED	3	703
Lexicographic, exact rank	PROVED	3	703
Hilbert, rational buckets	PROVED	3	703
Hilbert, budget 3	UNKNOWN (BUDGET)	0	3

The labels are *reported local outcomes*, not externally accepted proof-assistant certificates. The historical edition states that a second local checker replayed records, but independent validation of that program requires its code, inputs, and outputs. A source-reported test rejects a corrupted receipt; such a statement is not a substitute for reproducing the test.

A safe future release should distribute executable files, a pinned runtime and dependency list, deterministic inputs, raw proof traces, their exact checker specification, independent checked exports (if desired), and a command-line reproduction protocol. Finite-budget UNKNOWN is neither a refutation nor evidence of independence.

20 Question-driven research frontier and J4 boundary

The original research edition identified the following connected workstreams; this revision retains them as a scoped program, not as completed experiments:

1. Passive cancellation beyond the particular $E(G)$: classify active classes admitting sound projections.
2. Invariance when a controller interleaves commits, and its failure when candidate outcomes depend on within-layer updates.
3. Fairness, actual policy success, and restricted computable resource bounds.
4. Target-informed distributions under which a Hilbert or adaptive schedule can be demonstrably better than matched baselines.
5. Proof-certified history quotienting and minimal future-complete state representations.
6. Typed delegation across different calculi with separate verifiers and no unnoticed BANK widening.
7. Fair interleavings for infinite expanding action families without pretending to complete an infinite epoch.
8. Correlation of rational-address locality with proof-discovery usefulness.
9. A fully charged benchmark with wall-clock, memory, I/O, communication, and proof-custody costs.
10. Extension beyond the restricted set-theory fragment to accurately checked logical and ZFC axiom schemas with independently accepted proof exports.
11. A separately specified J4 physics model, if pursued, with frozen generators, assumptions, experimental geometry, and target equations. This paper provides no solution or physical evidence for J4.
12. Multiarticle comparisons: establish source consequence closures before crediting any synthesis with a truly transcendent theorem.

In particular, the supplied HFIS experiments concern navigation of a formally declared set-language search problem. They do not test a GR–QM deterministic/nondeterministic interface and do not solve the Bose–Oppenheim J4 obligation.

21 ATLAS v2.0 audit ledger and corrected claim statuses

ID	Concern and repair	Classification
H01	Distinguished nonempty positive-arity grid traversal from empty and zero-ary cases.	Corrected
H02	Added extra action-parameter caveat to tagged C^d formula.	Corrected
H03	Defined minimum *rule depth*; did not equate it with total proof length.	Corrected
H04	Restricted finite-Y breadth result to relative consequences.	Corrected
H05	Exposed infinite first-order schema obstacle to finite full-epoch hypothesis.	Corrected
H06	Supplied an alternate proof-level fair dovetailing theorem.	Conditional proof
H07	Distinguished existential winning path from actual policy's realized path.	Explicit counterexample
H08	Specified the passive GBG+ construction to ground cancellation and 2^n count.	Proved for model
H09	Repaired non-dominance proof using absolute rank sums.	Proved
H10	Marked historical toy results and Python tests as unreproduced without source code.	Evidence limited
H11	Scoped checker validity, BANK custody, active controller permissions, and resource charges.	Conditional
H12	Separated J4 and multiarticle research aspirations from demonstrated results.	Open

The proofs above establish claims under their respective explicit hypotheses. They do not independently validate all background cited papers or the source edition's companion implementation. A reviewer should not upgrade an open performance hypothesis to a theorem by replacing a difficult baseline with a weaker one.

22 Synthesis and next certification gate

Four different layers now have explicitly different roles: (1) formula syntax gives exact identity; (2) logical rules and verifier provide certificate legality; (3) Hilbert traversal assigns a schedule to a finite candidate geometry; and (4) GBG/GBG+ records complete authorized machine transitions. On the specified passive subclass, ordinary winning-state reachability is preserved. On a genuinely fair ongoing controller, finitary provability yields eventual verified discovery. On a finite frozen layer, Hilbert order preserves the completed consequence set. None of these implications supplies a universal speedup theorem, a finished ZFC prover, or an externally certified AMLD+0 settlement.

The release gate for stronger claims is concrete: independent reproduction of search traces; a public exact verifier or checked proof exports; explicit current run budgets and policy; a disclosed cost model; matched experiment controls; and separate publication of any stronger physics target, assumptions, and certificate. Until those gates are met, the supported result is that Hilbert geometry *can schedule* legal ATP work, not that it *improves* every ATP run.

References

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4. B. Tenneson, *Hilbert Space Filling Curve-Style Theorem Search: A Mathematical Specification, Constrained-Search Study, and Research Program*, version 0.5.0, August 2026.

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AI Disclaimer

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